Data Appendix

Predicting Wildfires from Time Series Weather Data

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1. Dataset Overview

This appendix documents the dataset used to predict wildfire occurrence in California using daily weather measurements. The dataset spans January 1, 1984 to January 10, 2025 and contains 14,976 observations after removing 12 rows with missing values.

Unit of Observation: Each row represents a single calendar day in California. It records weather conditions measured that day (precipitation, high/low temperature, wind speed) along with features derived from those measurements and a binary indicator of whether a wildfire started on that date.

Data Sources:

- National Oceanic and Atmospheric Administration (NOAA) Climate Data Online: provides the daily weather measurements.
- California Department of Forestry and Fire Protection (CAL FIRE): provides wildfire event records.
- The dataset was assembled and published on Zenodo.

2. Variable Descriptions

2.1 DATE

Type	date
Description	Date of the daily observation.
Variable Construction	No other variables were used to construct this variable.
N(m)	14976(0)

2.2 PRECIPITATION

Type	continuous (inches)
Description	Total precipitation recorded on that day.

Variable Construction	No other variables were used to construct this variable.
N(m)	14976(1)

2.3 MAX_TEMP

Type	continuous (degrees F)

Description	Maximum temperature recorded during the day.
Variable Construction	No other variables were used to construct this variable.
N(m)	14976(1)

2.4 MIN_TEMP

Type	continuous (degrees F)
Description	Minimum temperature recorded during the day.
Variable Construction	No other variables were used to construct this variable.
N(m)	14976(1)

2.5 AVG_WIND_SPEED

Type	continuous (mph)
Description	Average wind speed across the day.
Variable Construction	No other variables were used to construct this variable.
N(m)	14976(12)

2.6 FIRE_START_DAY

Type	binary (0/1)
Description	Indicates whether a wildfire started on this day. 0 = no wildfire; 1 = at least one fire started
Variable Construction	No other variables were used to construct this variable.
N(m)	14976(0)

2.7 YEAR

Type	integer
Description	Calendar year of the observation.
Variable Construction	This variable is derived from the DATE column.
N(m)	14976(0)

2.8 MONTH

Type	integer (1-12)
Description	Calendar month of the observation.

Variable Construction	This variable is derived from the DATE variable.
N(m)	14976(0)

2.9 TEMP_RANGE

Type	continuous (degrees F)
Description	Difference between MAX_TEMP and MIN_TEMP.
Variable Construction	This variable is derived from the MAX_TEMP and MIN_TEMP variables.
N(m)	14976(1)

2.10 WIND_TEMP_RATIO

Type	continuous (mph/F)
Description	Ratio of AVG_WIND_SPEED to MAX_TEMP.
Variable Construction	This variable is derived from the AVG_WIND_SPEED and MAX_TEMP variables.
N(m)	14976(12)

2.11 LAGGED_PRECIPITATION

Type	continuous (inches)
Description	Cumulative precipitation over the preceding 7 days.
Variable Construction	This variable is derived from the PRECIPITATION variable.
N(m)	14976(0)

2.12 LAGGED_AVG_WIND_SPEED

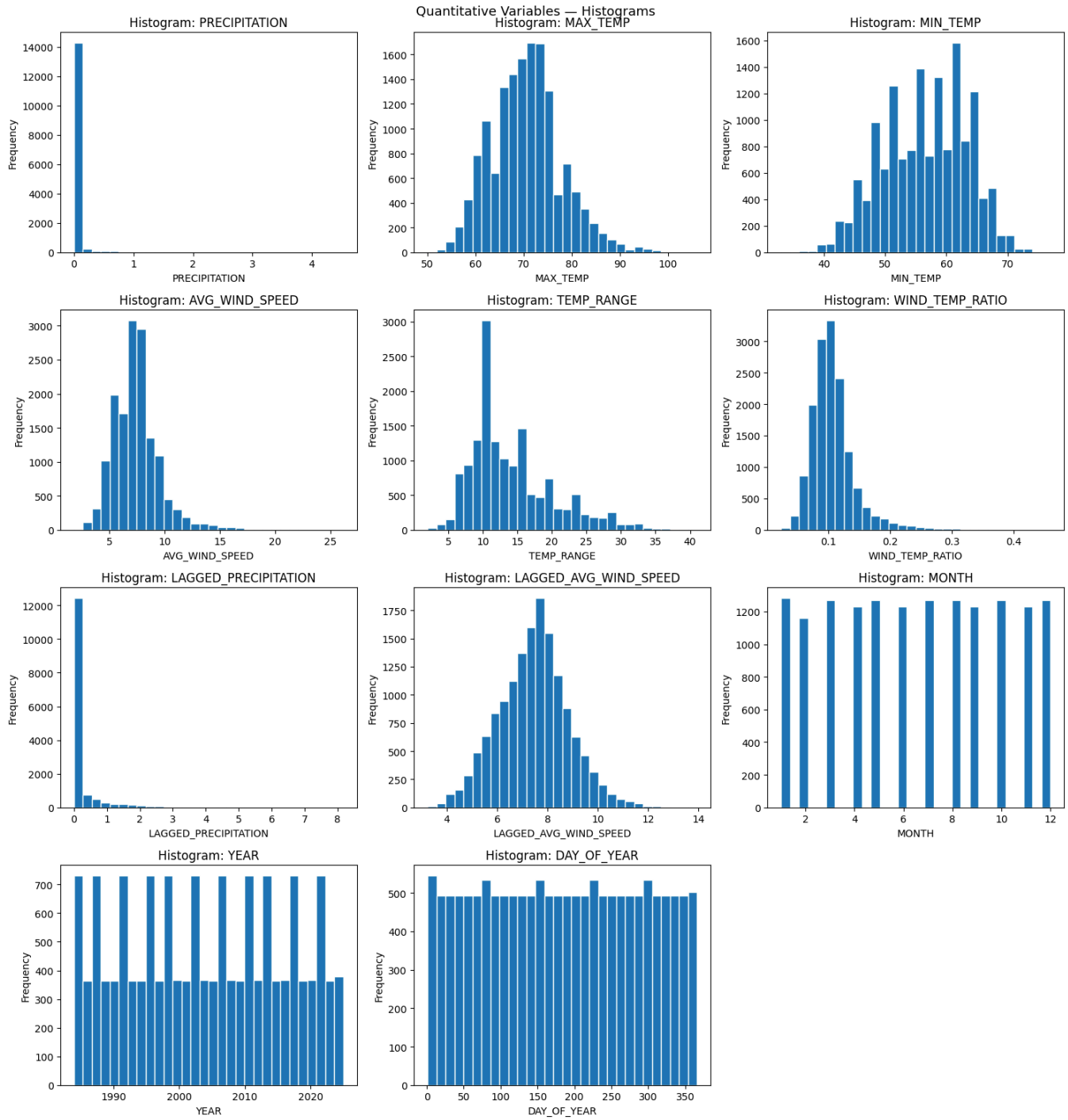
Type	continuous (mph)
Description	Average wind speed over the preceding 7 days.
Variable Construction	This variable is derived from the AVG_WIND_SPEED variable.
N(m)	14976(0)

2.12 DAY_OF_YEAR

Type	Integer (1-31)
Description	The day of the observation
Variable Construction	This variable is derived from the DATE variable.
N(m)	14976(0)

3. Quantitative Variable Histograms

The image below shows histograms for all quantitative variables in the dataset.



4. Summary Statistics

The table below presents descriptive statistics for all numeric variables in the analyzed dataset, excluding date variables (N = 14,976).

Statistic	PRECIP	MAX_T	MIN_T	AVG_WND	T_RANGE	WND/T	LAG_PRCP	LAG_WND	FIRE
count	14976.0000	14976.0000	14976.0000	14976.0000	14976.0000	14976.0000	14976.0000	14976.0000	14976.0000
mean	0.0323	70.5409	56.4997	7.4351	14.0413	0.1070	0.2264	7.4340	0.3319
std	0.1796	7.2620	6.7661	2.1300	5.9961	0.0356	0.6489	1.3850	0.4709
min	0.0000	50.0000	33.0000	1.7900	2.0000	0.0236	0.0000	3.2271	0.0000
25%	0.0000	65.0000	51.0000	6.0400	10.0000	0.0852	0.0000	6.5186	0.0000
50%	0.0000	70.0000	57.0000	7.1600	12.0000	0.1022	0.0000	7.4786	0.0000
75%	0.0000	75.0000	62.0000	8.5000	17.0000	0.1205	0.0600	8.2786	1.0000
max	4.5300	106.0000	77.0000	26.1700	41.0000	0.4591	8.1800	13.9329	1.0000